PRESIDENT'S OFFICE

REGIONAL ADMINISSTRATION AND LOCAL GOVERNMENT

ARUSHA REGIONAL EXAMINATION SYNDICATE

FORM THREE TERMINAL EXAMINATION MAY - 2021

CHEMISTRY - MARKING SCHEME

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1. ***10 Marks***

i	ii	iii	iv	v	vi	vii	viii	ix	x
A	C	A	D	D	A	C	B	B	A

2. ***5 Marks***

i	ii	iii	iv	V
E	G	B	D	A

3. (a) Required to write the ionic equations

i. $Pb(NO_3)_{2\,(aq)} + H_2SO_{4\,(aq)} \longrightarrow PbSO_{4\,(s)} + 2HNO_{3\,(aq)}$ ***1 Mark***

$Pb^{2+}_{(aq)} + 2NO_3^-{}_{(aq)} + 2H^+_{(aq)} + SO_4^{2-}{}_{(aq)} \longrightarrow PbSO_{4\,(s)} + 2H^+_{(aq)} + 2NO_3^-{}_{(aq)}$ ***1 Mark***

$Pb^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \longrightarrow PbSO_{4\,(s)}$ ***1 Mark***

ii. $Ca_{(s)} + 2HCl_{(aq)} \longrightarrow CaCl_{2\,(aq)} + H_{2\,(g)}$ ***1 Mark***

$Ca_{(s)} + 2H^+_{(aq)} + 2Cl^-_{(aq)} \longrightarrow Ca^{2+}_{(aq)} + 2Cl^-_{(aq)} + H_{2\,(g)}$ ***1 Mark***

$Ca_{(s)} + 2H^+_{(aq)} \longrightarrow Ca^{2+}_{(aq)} + H_{2\,(g)}$ ***1 Mark***

(b) Required any four importance of change of states of matter ***1 Mark @ = 4 Marks***

- **Water circle**: This is a continuous movement of water on, above and below the surface of the earth.
- **Refrigeration**: This uses the change of state to cause the cooling effect in refrigerators.
- **Refinery**: Refinery of petroleum and other liquids is possible only due to the evaporation process.
- **Metallurgy**: This involves the purification of metal from their ores and the manufacture of alloys.
- **Steam engines**: The liquid fuel of low pressure evaporates to make a ~~stem~~ steam of high pressure which is used to pu// a piston.
- **Drying material:** A material can dry faster if water evaporates easily.

4. (a) i. **Hard water:** is the water that does not form lather with soap.

While

Soft water: is the water that forms lather with soap. *2 Marks*

ii. **Temporary hardness of water** is caused by Hydrogen carbonate ion (HCO_3^-) *1 Mark*

Permanent hardness of water is caused by Sulphate ion (SO_4^{2-}) *1 Mark*

iii. Three methods for removing both temporary and permanent hardness of water are: *3 Marks*

- Distillation
- Addition of sodium carbonate/washing soda
- Use of ion exchange

(b) Given a hydrated magnesium chloride, $MgCl_2.6H_2O$

Total molecular mass = 1(24) +2(35.5) + 6(2 + 16)

= 203 *1 Mark*

Molecular mass of water = 6(2+16)

= 108 *1 Mark*

%Composition of water = $\frac{108}{203}$ x 100%

= 53.2% *1 Mark*

5. (a) i. **Empirical formula**: is the formula which represent the simplest ratio of the atoms in the molecule or compound.

1 Mark

Molecular formula: is the formula which shows the actual number of each atom in a molecule or compound.

1 Mark

(b) Given that

Total mass of iron oxide = 4.5g

Mass of iron = 3.15g *0.5 Mark*

Mass of oxygen = 1.35g *0.5 Mark*

Table *3 Marks*

Element symbol	Fe	O
Mass	3.15	1.35
Relative molecular mass	56	16
Mass/Relative molecular mass	0.056	0.084
Divide by smaller number	1	1.5
Whole ratio	2	3

Empirical formula is Fe_2O_3 *1 Mark*

(b) i. Balanced equation.

$$2Cu(NO_3)_{2\,(s)} \xrightarrow{\text{Heat}} 2CuO_{\,(s)} + 4NO_{2\,(g)} + O_{2\,(g)}$$ ***2 Marks***

ii. Decomposition reaction ***1 Mark***

6. (a) i. Water will combine with hot/burning metal to liberate hydrogen which is a flammable gas. Hydrogen will accelerate burning process. ***1 Mark***

ii. Because of a complete combustion of gas due to enough supply of air/oxygen. ***1 Mark***

iii. Because they have a complete filled outermost shells. ***1 Mark***

(b) (i) To complete the table ***1 Mark***

Table JJ

EXPERIMENT	PILOT	1	2	3
Final burette reading (cm^3)	16.30	16.00	32.00	48.30
Initial burette reading (cm^3)	0.00	0.00	16.00	32.00
Volume of acid used (cm^3)	**16.30**	**16.00**	**16.00**	**16.30**

Average volume of acid = $\frac{16.00+16.00+16.30}{3} = 16.10\ cm^3$ ***1 Mark***

(ii) Balanced chemical equation

$$2HCl(aq) + X_2CO_3(aq) \longrightarrow 2XCl(aq) + H_2O(l) + CO_2(g)$$ ***1 Mark***

Mole ratios: $n_a = 2, \quad n_b = 1$

(iii) Molarity of a metal carbonate

$$\frac{M_a v_a}{n_a} = \frac{M_b v_b}{n_b}$$

$$M_b = \frac{M_a v_a n_b}{V_b n_a}$$

$$= \frac{0.25 \times 16.1 \times 1}{25 \times 2}$$

Molarity of metal carbonate = $0.08 mol/dm^3$ ***1 Mark***

(b) (i) Required the relative atomic mass of a metal carbonate

$$\text{Concentration} = \frac{\text{Mass}}{\text{Volume of solution}} = \frac{2.12g}{0.25dm^3} = 8.48 g/dm^3$$

$$\text{Molar mass} = \frac{\text{Concentration}}{\text{Molarity}} = \frac{8.48 g/dm^3}{0.08 mol/dm^3} = 106 g/mol$$

Relative molecular mass = 106g/mol ***1 Mark***

(ii) Required the atomic mass of X.

$2(X) + 1(C) + 3(O) = 106$

$2X + 12 + 48 = 106$

$2X + 60 = 106$

$2X = 46$

$X = 23$

Atomic mass of X = 23g *1 Mark*

(iii) Required the electronic configuration of the metal ion

A metal is Sodium (Na)

A metal ion is Na^+

Electronic configuration of a metal ion (Na^+) is 2 : 8 *1 Mark*

7. (a) Four stages in hydrological cycle.

Evaporation: This is the process whereby water vapour/steam leaves the water bodies and goes into the air. *1 Mark*

Condensation: This is the process, in which water vapour in the atmosphere gets cold and change back to liquid,forming clouds. *1 Mark*

Precipitation: This occurs when so much water has condensed that the air cannot hold any more. The clouds get heavy and water falls back to the earth in the form of rain, hail and snow. *1 Mark*

Collection: When water fall back to the earth, some of it soaks to the ground and become part of ground water which plants and animals use. Some of water runs over the soil and collected in the rivers, lakes, streams, seas and ponds where cycle starts all over again. *1 Mark*

(b) **Periodicity**: is the recurrence of similar properties when elements are arranged in order of increasing their atomic masses. *1 Mark*

Modern periodic law states that "The properties of elements are periodic function of their atomic number" *1 Mark*

(c) Required to explain the variation of the following along a period in a periodic table.

i. **Electronegativity** increases from left to right due to increase in effective nuclear pull. *1 Mark*

ii. **Atomic size** decreases from left to right due to increase in effective nuclear pull. *1 Mark*

iii. **Ionization energy** increase due to the increase in effective nuclear pull. *1 Mark*

iv. **Non – metallic characters** decreases from left to right. *1 Mark*

8. (a) i. Gas **L** is Oxygen *1 Mark*

ii. Down ward displacement of water. *1 Mark*

Reason: Because oxygen is less dense than water and only slightly soluble in water. *1 Mark*

iii. Pure oxygen re – lights a glowing splint while ordinary air does not. *1 Mark*

(b) **Striking back of a flame** is the phenomena occurs when a flame passes down the barrel and burns at the air hole.

1 Mark

It is caused by:

- Opening of the air holes fully *0.5 Mark*
- Low gas pressure *0.5 Mark*

How to control:

- Turn off the gas *0.5 Mark*
- Close the air hole *0.5 Mark*
- Light the Bunsen burner again *0.5 Mark*

(c) RAM = $(\frac{69}{100} x\ 63) + (\frac{31}{100} x\ 65)$ *0.5 Mark*

= 43.47 + 20.15 *1 Mark*

= 63.62 *1 Mark*

9. (a) i. Ammonium chloride will sublimate leaving sand behind. *1 Mark*

ii. A cobalt (ii) chloride paper will change its colour from blue to pink. *1 Mark*

iii. The mixture will form a separate layer with cooking oil at the top and water at the bottom. *1 Mark*

iv. A hydrated copper (ii) sulphate (blue) will lose its water of crystallization forming anhydrous copper (ii) sulphate (white) *1 Mark*

(b) i. Ionic equation is an equation written in such a way that spectator ions are omitted. *1 Mark*

ii. Basicity of an acid is the number of hydrogen ions produced when one mole of the acid dissociates. *1 Mark*

iii. An indicator is a substance that changes from one colour in acid to another colour in base. *1 Mark*

(c) Any three characteristics of a good fuel *1 Mark @ = 3 Marks*

- It should have high heat content.
- It should be easily controlled.
- It should be cheap
- It should be easily stored and transported
- It should not produce dangerous by – products.
- It should have a high energy value.
- It should burn with a moderate speed.
- It should have a moderate ignition point

10. (a) Required to complete and balance

i. $MgO_{(s)} + H_2O_{(l)} \longrightarrow Mg(OH)_{2\,(aq)}$ *2 Marks*

ii. $2AgNO_{3\,(aq)} + MgCl_{2\,(aq)} \longrightarrow 2AgCl_{(s)} + Mg(NO_3)_{2\,(aq)}$ *2 Marks*

iii. $2SO_{2\,(g)} + O_{2\,(g)} \longrightarrow 2SO_{3\,(g)}$ *2 Marks*

iv. $NH_{3\,(g)} + HCl_{(aq)} \longrightarrow NH_4Cl_{(s)}$ *2 Marks*

(b) Consider the equation below

$2KClO_{3\,(s)} \xrightarrow{Heat} 2KCl_{(s)} + 3O_{2\,(g)}$

245g of $KClO_3 \longrightarrow 67.2\ dm^3$ of O_2

12.25g of $KClO_3 \longrightarrow$ V?

Volume of oxygen = $3.36\ dm^3$ *3 Marks*

(c) Required four importance of hard water

- Hard water has a good taste due to dissolved salts. *1 Mark*
- Calcium salts in hard water help in building strong teeth and bones *1 Mark*
- Hard water forms a coat of lime scale in pipes that prevent rusting and corrosion. *1 Mark*
- Hard water helps in formation of shells and egg shells in birds and other organisms. *1 Mark*

11. (a) Differences

i. **Strong acid** is an acid which dissociates completely in water to form a solution. *1 Mark*

While

Weak acid is an acid which dissociates partially in water to form a solution *1 Mark*

ii. **Concentrated acid** is an acid which has a strong affinity for water. *1 Mark*

While

Dilute acid is an acid that has a large amount of water *1 Mark*

(b) i. Solution C *1 Mark*

ii. Solution E *1 Mark*

iii. Solution B *1 Mark*

iv. Solution A *1 Mark*

(c) Applications of neutralization

Neutralization is the reaction between acid and base to form salt and water as the only products. This reaction h many important uses in our everyday life. Some of these include: *1 Mark*

Treating insect ~~strings~~: Bee and ant ~~strings~~ inject an acidic liquid to the skin or ~~flesh~~ which can be neutralized basic materials such as baking soda and cucumber. Wasp string injects basic liquid to the skin which can be neutralized by vinegar. *1 Mark*

Relieving indigestion: Indigestion is mostly caused by the presence of excess acid in the stomach. This can be treated by using ant acid medicines such as milk of magnesia or sodium bicarbonate. *1 Mark*

Management of soil pH: Various crops require different soil pH range for optimum growth and productivity. Excess soil acidity can be neutralized by addition of basic compounds such as oxides, hydroxides, carbonates and silicates Excess soil alkalinity can be lowered by addition of acidic substances such as sulphur, sulphate of iron and ammonium based fertilizers. *1 Mark*

Reducing acid rain: Acid rains are caused by gases such as carbon dioxide and oxides of sulphur present in the atmosphere. Most of these gases are emitted by factories and dissolve in rain water to form carbonic acid which cause air pollution. The air control devices known as scrubbers which are basic in nature are fitted to chimneys and exhaust pipes to neutralize the gas emitted. *1 Mark*

Treating factory wastes: Acidic liquid wastes from factories can kill aquatic organisms. To prevent this, such wastes are neutralized using calcium hydroxide before being released into the water bodies. *1 Mark*

Neutralize accidental spills: Acid spills in laboratory benches and floors can be neutralized by basic compound such as quick lime and slaked lime. Alkali spills can be neutralized by addition of hydrochloric acid. *1 Mark*